Daniel Batista da Silva

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Research Interests

Quantitative Asset Pricing; Volatility and Tail Risk; High-Frequency Finance; Portfolio Construction; Financial Econometrics

Education

Université de Genève – Swiss Finance Institute — Geneva, Switzerland
Ph.D. in Finance — 2021 – 2027 (expected)
Advisor: Olivier Scaillet

São Paulo School of Economics – FGV — São Paulo, Brazil
M.S. in Economics — 2019 – 2021
Thesis: Semivolatility Managed Portfolios
Advisor: Marcelo Fernandes

Federal University of Ceará — Fortaleza, Brazil
B.S. in Economics — 2015 – 2018
Thesis: Forward-looking behavior of Brazilian non-earmarked household credit
Advisor: Paulo Matos

Research Papers

The Jump–Diffusion Anatomy of Semibetas — *Work in Progress*

- Decomposes realized semibetas into continuous and jump components using high-frequency data to investigate the pricing of crash losses and insurance, and whether crash severity provides additional information about expected stock returns.

Semivolatility-Managed Portfolios — *Working Paper*
with Marcelo Fernandes — SSRN

- Develops portfolio strategies that jointly exploit upside and downside volatility, evaluates their risk-adjusted performance across equity factors, anomaly portfolios, and ETFs, and provides inference for Sortino-ratio comparisons.

Upside Risk and Return Timing in Bitcoin — *Revise & Resubmit, Economics Letters*
with Marcelo Fernandes — SSRN

- Shows that Bitcoin volatility spikes frequently reflect favorable upside states and that distinguishing upside from downside volatility improves risk-adjusted performance by avoiding indiscriminate deleveraging after positive volatility shocks.

Spectral Risk Factors and the Limits of Spanability — *Working Paper*

- Decomposes major risk factors into horizon-specific components and studies whether these risks can be represented and hedged by sparse portfolios of traded assets.

Conferences

2026 XXVI Brazilian Finance Meeting (EBFin), Fortaleza, Brazil
Presented: Semivolatility-Managed Portfolios; Upside Risk and Return Timing in Bitcoin; Spectral Risk Factors and the Limits of Spanability

2025 SFI PhD Workshop, Zurich, Switzerland
Presented: Spectral Risk Factors and the Limits of Spanability
Discussed: "Beyond Credit Spreads: What Loan Fund Flows Reveal about Real Activity?" by Wing Lam Cheung

2025 QMUL PhD Workshop, Queen Mary University of London, London, UK
Presented: Spectral Risk Factors and the Limits of Spanability

2025 São Paulo School of Advanced Science on High Dimensional Models (SPSAHDM), São Paulo, Brazil
Presented: Semivolatility-Managed Portfolios

2024 Annual Conference of the International Association for Applied Econometrics (IAAE), Thessaloniki, Greece
Presented: Semivolatility-Managed Portfolios

2024 SFI Research Days, Study Center Gerzensee, Switzerland
Presented: Semivolatility-Managed Portfolios
Discussed: "What Is the Consumption–Wealth Return?" by Paul Schneider, Marc Van Uffelen, and Paul Whelan

2023 Financial Econometrics Meets Machine Learning (FinEML), Rotterdam, Netherlands
Presented: Semivolatility-Managed Portfolios

2023 IFABS 2023 Oxford, Oxford, UK
Presented: Semivolatility-Managed Portfolios

2023 FMA European Conference, Aalborg, Denmark
Presented: Semivolatility-Managed Portfolios
Discussed: "Which Is Worse: Heavy Tails or Volatility Clusters?" by Joshua Traut and Wolfgang Schadner

Research Experience

Ph.D. Researcher – Asset Pricing and Financial Econometrics 2021 – Present

- Conduct empirical research using high-frequency, time-series, and large cross-sectional financial datasets.
- Develop reproducible research pipelines for data processing, econometric estimation, portfolio construction, and backtesting.
- Implement statistical inference, simulation, robustness, and out-of-sample analyses primarily in R and Python.

Industry Experience

B3 – Brasil, Bolsa, Balcão São Paulo, Brazil
Risk Analyst II, Risk Parameters Modeling Division 2020 – 2021

- Developed and validated market-risk parameters used in B3's CORE central counterparty risk model.
- Estimated and proposed initial risk parameters for new products, including IPOs and newly listed ETFs, supporting pre-launch calibration and Risk Committee approval.
- Monitored parameter stability and risk metrics under normal and stressed market conditions.

Intern, Risk Parameters Modeling Division 2019 – 2020

Teaching Experience

University of Geneva 2022 – Present
Teaching Assistant: Financial Econometrics; Stochastic Processes in Finance; Quantitative Risk Management
Instructor: Python short courses for finance and statistics

São Paulo School of Economics – FGV 2019 – 2020
Teaching Assistant: Introduction to Stochastic Processes; Econometrics of Financial Series
Instructor: R short courses

Federal University of Ceará (UFC) 2016 – 2018
Head Teaching Assistant: Calculus; Statistics; Econometrics

Refereeing

Ad Hoc Referee, Brazilian Review of Finance 2021
Ad Hoc Referee, Spanish Journal of Finance and Accounting 2026

Skills

Programming: R (advanced), Python (advanced), MATLAB (intermediate)
Methods: asset pricing; financial econometrics; stochastic processes; realized measures; volatility and jump models; portfolio optimization; statistical inference; Monte Carlo simulation
Data: time-series data; high-frequency financial data; panel data; large cross-sectional datasets
Tools: LaTeX, Git, Linux

Languages

Portuguese (native), English (fluent), French (advanced), Spanish (intermediate)